

JIM LOWRY

PROFESSIONAL SUMMARY

Versatile technical professional and AI solutions architect with 10+ years of Microsoft platform expertise, specializing in AI-assisted business solution development and enterprise project management. Successfully architected and deployed three commercial-grade platforms through strategic AI collaboration, demonstrating advanced prompt engineering and human-AI partnership capabilities. Proven ability to translate complex business requirements into scalable technical solutions by orchestrating AI-generated development while ensuring enterprise-grade quality and integration.

CORE COMPETENCIES

- AI Solutions Architecture & Multi-Agent Systems
- Advanced AI Prompt Engineering & Human-AI Collaboration
- Microsoft Power Platform (Dataverse, Power BI, Power Apps, Power Automate)
- AI-Assisted Full-Stack Development and Orchestration
- AI-Guided Enterprise Security & Cloud Implementation
- Strategic AI Collaboration & Solution Architecture
- Predictive Analytics & Machine Learning Implementation
- Business Intelligence & Data Visualization
- Blockchain Technology Integration
- Cross-functional Team Leadership & Stakeholder Management

PROFESSIONAL EXPERIENCE

Self-Employed – High Ridge, MO

AI Solutions Developer & Full-Stack Engineer 3/2025 – Present

- Architected enterprise-grade multi-agent AI platform with 5 specialized business intelligence agents coordinating through custom Model Context Protocol (MCP) implementation
- Implemented Google Cloud KMS enterprise security migration, replacing legacy encryption with hardware-backed AEAD encryption and managing 36+ API endpoint migrations
- Developed sophisticated AI workflow orchestration system using n8n automation platform with dynamic agent coordination and intelligent session management
- Integrated professional SEO intelligence APIs including DataForSEO, Google Analytics 4, and Search Console for autonomous competitive analysis and strategic recommendations

Note: All development projects executed through strategic AI collaboration with direct oversight of architecture, requirements, and quality assurance

AI Art Authentication Platform (Airtisan.net)

- Designed and developed React-based blockchain certification platform targeting high-value art collectors with AI computer vision integration concepts
- Implemented conversion-optimized landing pages with A/B testing capabilities and comprehensive SEO strategy including structured data (JSON-LD) and semantic HTML
- Created sophisticated user experience flows for AI authentication processes with Firebase backend integration and multi-factor authentication
- Built scalable application architecture following React best practices with context-based state management and component-driven design.

Workforce Analytics Platform (Lowrys.org)

- Built predictive analytics application using Next.js 15 and TypeScript, processing 50,000+ Bureau of Labor Statistics occupation records with ML-based attrition forecasting.
- Implemented real-time data visualization using Chart.js, React-ChartJS-2, and React-Globe.gl for 3D geographic data representation
- Developed automated ETL pipelines for government dataset processing with 99%+ accuracy and intelligent data normalization
- Created comprehensive business intelligence framework with 65+ encrypted strategic data fields and cross-agent data sharing capabilities.

AI Collaboration & Development Innovation

- Pioneered advanced AI-assisted development workflows demonstrating strategic prompt engineering and multi-model AI coordination (Claude Sonnet, GPT-4 Turbo)
- Successfully identified and bridged AI limitations through strategic problem decomposition, quality assurance, and integration oversight
- Established efficient human-AI collaboration methodologies for complex enterprise-grade application development

Ameren – St. Louis, MO

Data Analyst, Project Manager & Data Systems Developer 5/2016 – 3/2025

- Advanced from PMO Analyst to Project Manager to PMO Data Systems and Reports Developer
- Developed and maintained business intelligence solutions integrating data from multiple sources into comprehensive reporting dashboards, serving over 200+ PMO end users
- Utilized data analytics to management insights like cross-team dependencies across 100+ projects, enabling advanced resource planning and optimization of support staff allocation, resulting in improved project delivery timelines and resource utilization.
- Created automated reports and dashboards improving data accessibility and decision-making
- Collaborated with cross-functional teams to gather requirements and deliver customized business intelligence solutions
- Implemented data modeling best practices and ETL processes to ensure data accuracy and reliability
- Provided training and support to end users on Power BI tools and dashboard functionality
- Delivered training and knowledge transfer to teams on BI tools and analytical methodologies
- Managed technology projects and cybersecurity deployments for utility plant operations
- Performed business analysis duties including requirements gathering, process documentation, and operations analytics
- Provided 3rd party consulting services for operations analytics and business intelligence testing in simulation environments, testing and implementation
- Designed and developed reporting systems and analytical tools to support strategic business decisions
- Developed project plans and coordinated with cross-functional teams to ensure successful delivery
- Established project metrics and performance indicators to track objectives and deliverables

CSX Transportation - St. Louis, MO

Signalman, Conductor, Remote Train Operator 6/1998 – 5/2014

- Implemented advanced signaling systems across multiple states, maintaining FRA required safety and compliance standards.
- Manage train operations within FRA requirements and standards.

NOTABLE PROJECTS & PLATFORMS

AI SEO Strategies Platform (SEO.Lowrys.org)

- Enterprise multi-agent AI platform with autonomous SEO optimization and commercial pricing tiers
- 5-agent coordinated architecture with Google Cloud KMS security and n8n workflow automation
- Professional SEO intelligence integration with DataForSEO, Google Analytics 4, and Search Console APIs
- Advanced authentication system with Firebase integration and role-based access control

AIrtisan Art Authentication (AIrtisan.net)

- Blockchain-based art authentication platform for high-value collectors and investors
- React-based application with Firebase backend and conversion-optimized user experience
- AI computer vision integration concepts for artwork verification and provenance tracking
- Mobile-responsive design with accessibility compliance (ARIA) and technical SEO implementation

Workforce Analytics Platform (Lowrys.org)

- Predictive analytics platform processing 50,000+ government occupation records
- ML-based attrition forecasting with real-time data visualization and 3D geographic representation
- Next.js 15 with TypeScript, automated BLS data integration, and comprehensive ETL pipeline processing
- Professional portfolio showcasing data analytics and technical expertise

EDUCATION

Bachelor of Science in Information Management

Webster University | Webster Groves, St. Louis, MO | 2017

TECHNICAL SKILLS

AI Collaboration & Solution Architecture:

- Advanced AI Prompt Engineering & Strategic Collaboration
- Complex Problem Decomposition for AI Implementation
- AI-Assisted Development Workflow Orchestration
- Multi-Model AI Coordination (Claude Sonnet, GPT-4 Turbo)
- AI Limitation Analysis & Gap Bridging
- Quality Assurance & Integration Oversight
- Business Requirements Translation to AI Implementation

AI & Machine Learning:

- Multi-Agent AI Systems
- Anthropic Claude Sonnet
- OpenAI GPT-4 Turbo
- Model Context Protocol (MCP)
- AI Workflow Orchestration
- Predictive Analytics & Statistical Modeling
- Machine Learning Implementation

Enterprise Security & Encryption:

- Google Cloud KMS
- AES-256-GCM Encryption
- Multi-Factor Authentication (MFA)
- Enterprise Security Architecture
- JWT Token Management
- Firebase Authentication

- Role-Based Access Control

Modern Web Development:

- Next.js 14/15 (App Router, Server-Side Rendering)
- React 18 (Hooks, Context API, Concurrent Features)
- TypeScript 5.0
- JavaScript (ES6+)
- Tailwind CSS
- Node.js & API Development
- Responsive Web Design
- Component-Based Architecture

Cloud & Infrastructure:

- Google Cloud Platform (KMS, Service Accounts)
- Vercel (Serverless, Edge Computing)
- Firebase (Authentication, Firestore, Cloud Functions, Messaging)
- Turso/LibSQL (Distributed Edge Database)
- n8n (Workflow Automation)
- Webhook Architecture
- Microservices Architecture

Data Visualization & Analytics:

- Chart.js & React-ChartJS-2
- React-Globe.gl (3D Geographic Visualization)
- Predictive Modeling & Forecasting
- Business Intelligence Dashboards
- Real-time Analytics Implementation
- Statistical Analysis & Trend Forecasting

Microsoft Power Platform:

- Dataverse
- Power BI
- Power Apps
- Power Automate
- Dataflows
- Power Query (M Language)

Database Technologies:

- SQL
- SQLite/LibSQL
- Firestore (NoSQL)
- Access
- Dataverse
- Data Modeling
- Database Design and ERDs
- Complex Multi-Table Relationships
- Encrypted Field Management
- Database Migration Strategies
- Cross-Agent Data Sharing

API Integration & Data Processing:

- DataForSEO API
- Google Analytics 4 & Search Console APIs
- BLS (Bureau of Labor Statistics) API
- Stripe Payment Processing
- RESTful API Development
- ETL Pipeline Development
- Automated Data Processing
- Data Validation & Quality Assurance

Microsoft Office Suite:

- Microsoft 365
- Advanced Excel
- Word
- PowerPoint
- Outlook
- Access
- OneDrive

Direct Technical Expertise:

- VBA, DAX, M (Power Query) - Advanced proficiency
- SQL - Business intelligence and reporting
- Microsoft Power Platform - Expert level
- Advanced Excel - Complex formulas and automation

AI-Assisted Development Capabilities:

- TypeScript/JavaScript (AI-generated with strategic oversight)
- Python (AI-implemented with business logic guidance)
- Complex database queries (AI-generated with requirements specification)

Blockchain & Emerging Technologies:

- Blockchain Certification Concepts
- Digital Provenance Systems
- Cryptocurrency Integration Concepts
- Decentralized Application Architecture

Business Intelligence:

- ETL Processes
- Dashboard Development
- Data Visualization
- Reporting
- Performance Analytics
- KPI Tracking

Project Management:

- Agile
- Waterfall
- SDLC
- Process Improvement
- Stakeholder Management

- Requirements Gathering
- Cross-functional Team Leadership

SEO & Digital Marketing:

- Technical SEO Implementation
- Structured Data (Schema.org, JSON-LD)
- Meta Optimization
- Semantic HTML
- Mobile-Responsive Design
- Conversion Optimization
- Analytics Implementation

MILITARY TRAINING & CERTIFICATIONS

- U.S. Navy | Interior Communications Electrician Training
- MECP Expert Installation Technician Certification

ADDITIONAL INFORMATION

- Advanced AI collaboration and prompt engineering expertise with proven commercial application success
- Strong analytical and problem-solving abilities with attention to detail and accuracy
- Excellent communication and presentation skills with experience presenting to executive leadership
- Experience with cross-functional team collaboration and stakeholder management
- Proven track record in delivering data-driven business solutions on time and within budget
- Ability to translate complex technical concepts into business-friendly language
- Continuous learner staying current with emerging technologies and industry best practices
- Demonstrated ability to architect and deploy commercial-grade applications across multiple domains
- Expertise in bridging AI limitations through strategic problem decomposition and quality assurance